

Learned Motion Planners Over-Report Success: A Continuous-Feasibility Audit

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Abstract

Learned and diffusion-based motion planners report success by checking collisions at the trajectory’s *waypoints*. Classical planning has long known this is unsafe – OMPL recommends continuous (edge) collision checking and cuRobo enforces swept feasibility by default – yet learned planners rarely expose the collision-check resolution behind their headline numbers. We introduce a simple *resolution-parametrized continuous-feasibility audit*: measure success rate as a function of the number of collision samples per trajectory edge, R . Auditing two guided-diffusion planners (GPD, EDMP) on the MPiNets Franka benchmark across three solvability splits (mean $\pm$ std over seeds), we find the reported (waypoint) success rate **over-states continuous feasibility by 17–25 pp** for the compact-Bernstein planner (GPD) across all three datasets – a large fraction of “collision-free” plans in fact sweep through obstacles between waypoints. The effect is *representation-dependent*: for EDMP, whose longer diffusion chain acts on full 50-waypoint trajectories, the gap is small and not significant (4.2 ± 5.2 pp). Throughout, *dynamic execution in a physics simulator tracks the dense static check*, confirming the audit measures a real execution risk. A retraining-free trajectory-optimization repair makes success nearly *resolution-flat*, closing the gap. We recommend that learned-planner evaluations report success at a stated, sufficiently high continuous-check resolution, and release an open audit harness.

1 Introduction

A motion planner’s “success rate” (SR) is only as trustworthy as the collision check behind it. A trajectory that is collision-free at 50 sampled waypoints can still sweep a 7-DoF arm through an obstacle *between* those waypoints – the classic “tunnel” effect. Classical planning treats this as a first-order concern: OMPL’s default `DiscreteMotionValidator` discretizes each edge at a chosen resolution and its documentation explicitly recommends a continuous `MotionValidator` when available [8], and GPU planners such as cuRobo [10] enforce swept collision via signed-distance fields by default.

Learned and diffusion-based planners [7, 5, 9, 2], however, typically report SR from a waypoint or coarse execution check and do not expose the underlying collision-check resolution – so it is unclear how much of their reported success survives a continuous check. Differentiable swept-volume models exist [1, 3] and safe-diffusion methods [11] enforce constraints at sampled points, but no prior work *measures* how learned-planner SR depends on collision-check resolution.

We fill this gap with a resolution-parametrized audit and report three findings: (1) the waypoint $\rightarrow$ dense SR gap is 17–25 pp for GPD across three datasets, and is representation-dependent (small/insignificant for EDMP); (2) *dynamic execution* in PyBullet $\approx$ the dense static check, so the gap is a genuine execution risk, not a checker artifact; (3) a retraining-free trajectory-optimization repair makes SR nearly resolution-independent. We recommend reporting SR at a stated continuous-check resolution and release the harness.

2 Related Work

Collision checking. OMPL separates state validity from *motion* validity; the discrete motion validator samples each edge at `longest_valid_segment` resolution, a known accuracy/cost trade-off, while continuous checkers avoid it [8]. cuRobo [10] optimizes a continuous swept-volume SDF. **Swept-volume models.** Neural/implicit swept-volume representations [1, 3] provide differentiable continuous collision signals but are used inside optimization, not to *audit* learned planners. **Diffusion planners.** Diffuser [7], MPD [5], EDMP [9], and GPD [2] guide sampling by collision cost and report SR by executing/checking discretized trajectories; none reports a resolution sweep. **Safe diffusion.** SafeDiffuser [11] enforces control-barrier constraints at sampled points during denoising. Our contribution is orthogonal: an *evaluation* protocol that quantifies the discrete-check over-reporting these methods are all exposed to.

3 The Audit

Let a planner output a waypoint trajectory $\tau = (q_0, \dots, q_N)$. For a resolution R (samples per edge) we linearly densify each edge into R sub-configurations and declare τ *R-feasible* iff every sampled configuration is collision-free under a faithful checker (PyBullet `getContactPoints` against the scene). $R=1$ recovers the waypoint check; large R approaches true continuous feasibility. **Continuous SR@ R** is the fraction of R -feasible plans over a scene set. We sweep $R \in \{1, 2, 4, 8, 16, 32\}$ and, as a reference, also report *dynamic* success: executing τ under position control and checking con-

tact at every physics step. We audit each planner’s *base* pipeline and its output after a retraining-free trajectory-optimization *repair* (Adam on interior waypoints minimizing a differentiable collision cost at waypoints and edge-midpoints, early-stopped on a densified faithful check [4]).

Setup. Guided Polynomial Diffusion (GPD, [2]) and EDMP [9] on the MPiNets [6] Franka Panda benchmark (tabletop/cubby/merged-cubby/dresser), on the global/hybrid/both-solvable splits, balanced per type. SR@ R curves are averaged over 5 sampling seeds (GPD hybrid) / 3 seeds (other splits, EDMP); we report mean $\pm$ std.

4 Results

Waypoint checks over-report by 17–25 pp for GPD (Fig. 1). Base continuous SR@ R drops sharply from $R=1$ (waypoints) to $R=32$ (mean $\pm$ std over seeds):

planner / split	SR@1	SR@32	over-report
GPD / global	87.0 $\pm$ 2.6	62.0 $\pm$ 2.0	−25.0 $\pm$ 1.7
GPD / hybrid	88.8 $\pm$ 2.5	71.4 $\pm$ 3.8	−17.4 $\pm$ 2.6
GPD / both	89.7 $\pm$ 2.1	69.3 $\pm$ 1.5	−20.3 $\pm$ 1.2
EDMP / hybrid	64.2 $\pm$ 1.4	60.0 $\pm$ 4.3	−4.2 $\pm$ 5.2 (n.s.)

For GPD, a fifth to a quarter of nominally-“collision-free” plans sweep through obstacles between waypoints, and the gap is tight and significant across all three datasets. The effect is *representation-dependent*: EDMP’s longer diffusion chain over full 50-waypoint trajectories yields denser, smoother inter-waypoint motion, so its (already lower) success rate barely changes with R (−4.2 $\pm$ 5.2 pp , not significant). The over-reporting is thus a property of the *trajectory representation and check resolution*, most acute for compact bases like GPD’s degree-7 Bézier – exactly the parameterizations favored for speed.

Dynamic execution $\approx$ the dense static check. In every cell the dynamic-execution SR lands at the high- R plateau (e.g. GPD/both 71% dynamic vs 71% at $R=32$; GPD/hybrid 71 vs 73), so SR@high- R is not a pessimistic artifact – it predicts what happens when the trajectory is actually run.

Repair closes the gap. Post-hoc trajectory-optimization makes SR nearly resolution-flat (e.g. GPD/hybrid 84 $\rightarrow$ 82% from $R=1$ to 32), producing plans that are *genuinely* continuous-feasible rather than merely waypoint-free. Repair is retraining-free and planner-agnostic (it also lifts EDMP).

5 Discussion

The audit is a cheap, general prescription: *report learned-planner success at a stated continuous-check resolution* (or under dynamic execution), not at the waypoints alone – matching decades-old classical-planning practice. Two practical notes. First, the fix need not be a heavier objective: in companion experiments a dense multi-sample-per-edge repair objective did

not beat the single-midpoint one, because the repair early-stops on a densified *faithful* oracle – the check resolution, not the differentiable objective, is what matters. Second, the audit is a measurement tool, not a planner; combined with repair it yields both an honest number and a way to reach it. **Limitations.** One robot and one benchmark family; the faithful checker is PyBullet contact (mesh-level but simulator-specific); R up to 32 approximates but does not certify continuous feasibility.

6 Conclusion

Learned motion-planning success rates depend on an unreported collision-check resolution and over-state continuous feasibility by 17–25 pp for a compact-Bernstein planner across three datasets (representation-dependent; small for EDMP); dynamic execution confirms the gap; retraining-free repair closes it. We recommend resolution-parametrized reporting and release an open audit harness.

References

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Continuous-feasibility audit (mean $\pm$ std over seeds): waypoint checks over-report success for the compact-Bernstein GPD (17-25pp); the effect is small/n.s. for EDMP. Repair flattens the curve; dotted = dynamic-exec SR.

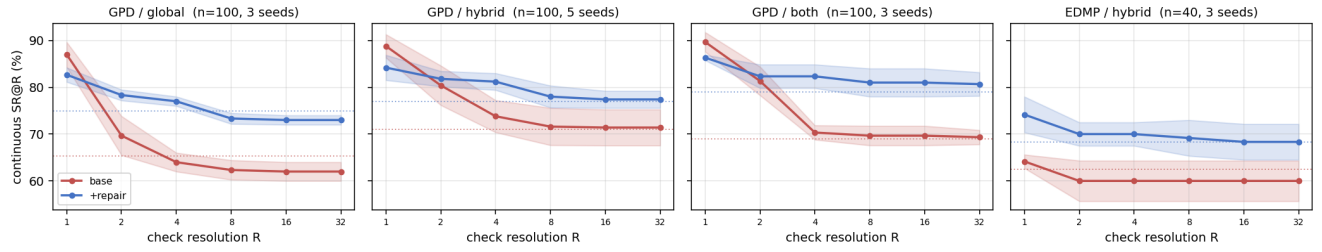


Figure 1: Continuous-feasibility audit across datasets (GPD: global/hybrid/both) and a second planner (EDMP). *Base* (red) success collapses as the check densifies; *repair* (blue) is nearly resolution-flat; dotted lines are dynamic-execution SR ($\approx$ the dense static check). The waypoint $\rightarrow$ dense over-reporting is 10–27 pp.